



Fw: Questions re. dead-reckoning paper

From Jojo Schultz <jojo.schultz@outlook.de>

Date Mon 20/04/2026 10:59

To Oakleigh Wilson <Oakleigh.Wilson@research.usc.edu.au>

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From: Gunner, Richard <rgunner@ab.mpg.de>

Sent: Saturday, December 7, 2024 11:17:43 PM

To: Jojo Schultz <jojo.schultz@outlook.de>

Subject: Re: Questions re. dead-reckoning paper

Hi Jojo,

The short answer is, yes - if a wrong ref frame was chosen then DR path will be inaccurate. But comparing subsets with GPS pre-correction will help debug if chosen orientation was correct. The patterns and directions of travel should be similar even if scale is not when compared to GPS. I had some lions where I had to 'figure' out the orientation post release. But visually it should be easy to decipher the heave axis of movement in acc and mag (acc will be the one closest to +/-1 g most of the time, and mag will be the one changing the least)... In terms of deciphering the acc/mag surge and sway dimensions, this may be more challenging. But for the acc, you would expect greater variation in surge dimension because they stick their head to ground to forage and then stick their head back up when travelling and vigilance, so that may help. If mag is different orientation set up to acc, then it's worth just checking if you switch the surge/sway orientations then which one produces most realistic DR track relative to GPS... by deciphering the heave axis and dimension of measured g, you already have homed in on just a few possible orientations it could be...

Best,

Rich

From: Jojo Schultz <jojo.schultz@outlook.de>

Sent: 04 December 2024 06:26:33

To: Gunner, Richard

Subject: Re: Questions re. dead-reckoning paper

Hi Richard,

I am sorry this took me a while to answer you!

I had a thorough look through your debugging and extracted some ideas and also thought of some further steps we may need to incorporate in our data.

Thank you so much for taking the time to go through the rugby field test so thoroughly!! This is amazing!

I was wondering: When you put collars on the animals and release them, so you know the exact orientation and heading when the animals wake up and head off and do you use these then as the acc.ref.frame / mag.ref.frame? I guess the key issue in the debugging was exactly that - however the team who collected the African Impala data did not note down the release orientation. Hope this question makes sense.

I am still not a complete expert with DR, but in my understanding it would change the DR path if the wrong ref frame was chosen. With the rugby field it was quite easy to see as we knew exactly the correct track that we walked and I can see how you then chose the NWU ref frame for the magnetometer, but for a wild animal apart from the GPS fixes that is unknown.

And one more mini question: There is so much data when collecting over multiple weeks - do you usually work on a "preview" of a subset to make sure everything is correct before handling all of it?? I was thinking to write a loop that makes chunks first, cleans the GPS data and then does DR on a subset to check the orientations etc. before looking at the entire collar data.

Sorry to ask more questions 😊 It's a cool project, Chris however has little funding to actually have me work on it now, so I'm just doing a bit more in my free time out of interest. Hopefully I can get this as far as I can and show you a snippet soon 😊

Hope otherwise everything is going well!

Kind regards,
Jojo

From: Gunner, Richard <rgunner@ab.mpg.de>
Sent: Wednesday, 6 November 2024 02:11
To: Jojo Schultz <Jojo.Schultz@outlook.de>; Luke Jessup <luke.jessup@uq.edu.au>
Cc: Christofer Clemente <cclement@usc.edu.au>
Subject: Re: Questions re. dead-reckoning paper

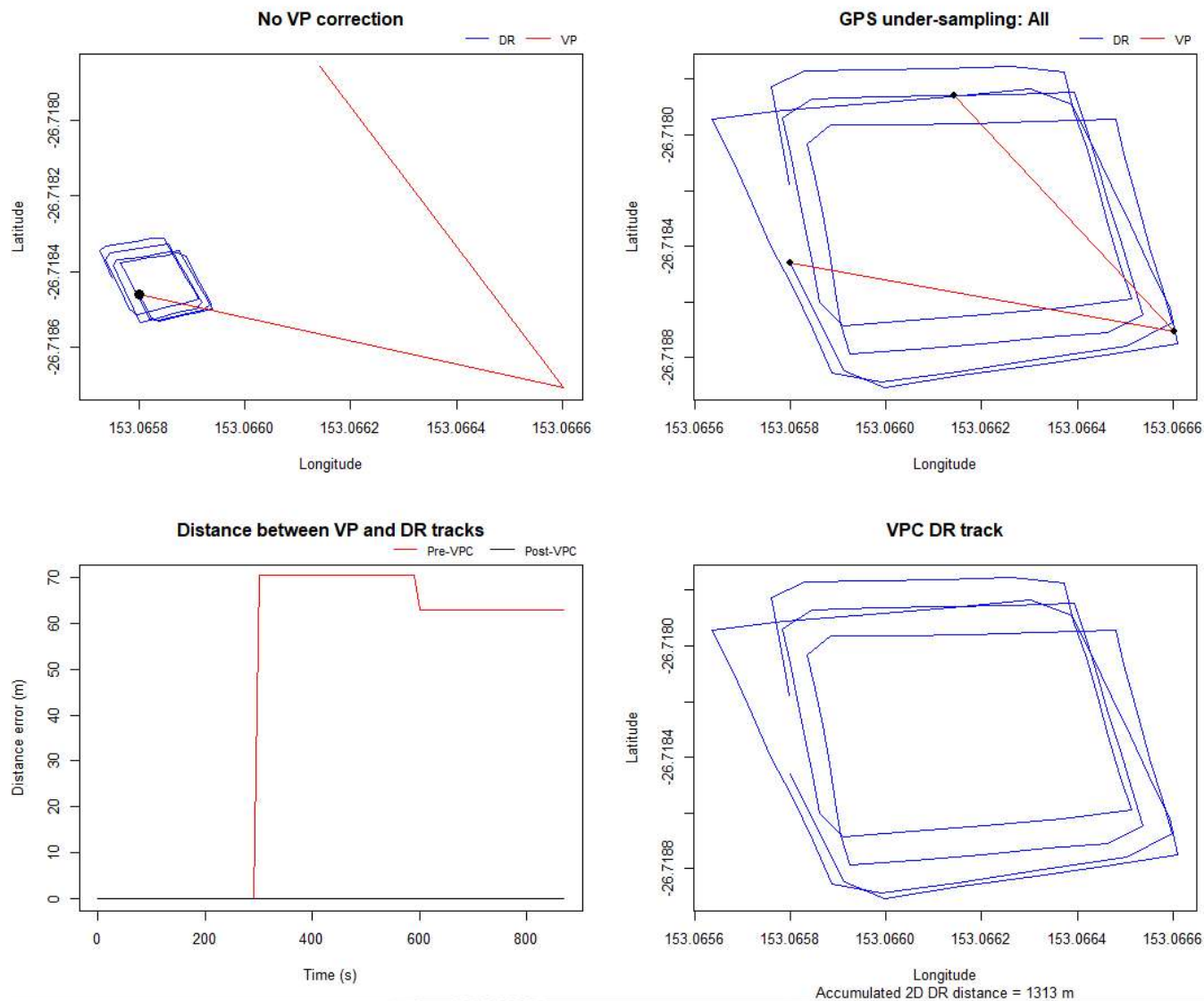
Hi Jojo,

Everything is working as expected, so no need to worry! The issue turned out to be quite simple—the magnetic coordinate system is arranged slightly differently (in an orthogonal layout) compared to the accelerometer. In Gundog.Compass, setting the following parameters resolves everything:

acc.ref.frame = "NED"
positive.g = "up"
mag.ref.frame = "NWU"

I've attached an R script that provides a complete walkthrough of my investigation, which might be helpful. I've also included a short PowerPoint presentation with the results.

In summary, applying a pitch offset to account for the device being tilted downward and calibrating with a few GPS points during the walking period yielded these results: (though even without applying pitch offset, DR track remained similar, which should be the case because during heading computation, the vectors are de-rotated according to the normalised g offset):



From: Gunner, Richard

Sent: 05 November 2024 12:14:14

To: Jojo Schultz; Luke Jessup

Cc: Christofer Clemente

Subject: Re: Questions re. dead-reckoning paper

Hi Jojo,

I got a bit bogged down with some deadlines in the last week, but I haven't forgotten about this... I have some time in the next couple days, so results coming soon 😊

Best,

Rich

From: Jojo Schultz <jojo.schultz@outlook.de>

Sent: 04 November 2024 05:00:40

To: Gunner, Richard; Luke Jessup

Cc: Christofer Clemente

Subject: Re: Questions re. dead-reckoning paper

Hi Richard,

With Luke now in the US, I have taken over this project and am currently working myself into the code and the theory behind dead-reckoning.

I have started a massive code diagram too.

Anyway, I was just wondering if you had a chance yet to have a look at this rugby field test or if you need anything else from our end.

Thanks, hope all is going well!

Jojo

From: Gunner, Richard <rgunner@ab.mpg.de>

Sent: Monday, October 21, 2024 20:36

To: Luke Jessup <luke.jessup@uq.edu.au>

Cc: Christofer Clemente <cclement@usc.edu.au>; Jojo Schultz <Jojo.Schultz@outlook.de>

Subject: Re: Questions re. dead-reckoning paper

Thanks Luke,

I'll let you know how I get on in the interim. Good luck with your move!

Best,

Rich

From: Luke Jessup <luke.jessup@uq.edu.au>

Sent: 21 October 2024 12:30:08

To: Gunner, Richard

Cc: Christofer Clemente; Jojo Schultz

Subject: RE: Questions re. dead-reckoning paper

Hi Richard,

I collected some data today, walking around a rugby field – 4 laps per condition. One condition involved walking with the device held ~flat, and the other with the device pitched ~45 deg (image attached). If you follow this link <https://drive.google.com/drive/folders/1FWBkPoeex4uaUpCahAsV0QAXS4f47gJa?usp=sharing> I've uploaded a folder containing both the IMU and GPS data synced in real-time (IMUGPSdf.txt), as well as the separate raw IMU (dataLog.txt) and GPS (serialLog.txt) data files, and IMU calibration data file. I've also attached an image that shows the direction of each axis, and an image that shows the path that I took (start / end point indicated by the star). Note: I tried to keep a consistent path but might have drifted at times because there were no line markings on the field; the data files contain quaternion data as Q9_1, Q9_2 and Q9_3; for the raw files, ~2048 LSB = 1 G, and the magnetometer data has a sensitivity scale factor of 0.15 (<https://invensense.tdk.com/wp-content/uploads/2024/03/DS-000189-ICM-20948-v1.6.pdf>); and IMU data collected at 50 Hz, GPS data every 5 min. Please let me know if I can provide any further details!

I ran the data through our current dead reckoning pipeline using the accelerometer and magnetometer data. If you get the same results as me, you'll notice that the dead reckoning works great when the device is oriented flat (i.e. overlapping rectangles), but that changes in heading are underestimated when the device isn't flat (i.e. zig-zagging) – image attached.

Thanks so much for your help with this!! 🙏 I'll try to be as responsive as possible; I move to the US on Monday next week to start my new position. Jojo (Cc'd in) will be taking charge of this project for us.

Best wishes,
Luke

From: Luke Jessup
Sent: Saturday, October 12, 2024 7:48 AM
To: Gunner, Richard <rgunner@ab.mpg.de>
Cc: Christofer Clemente <cclement@usc.edu.au>
Subject: RE: Questions re. dead-reckoning paper

Hi Rich,

No worries haha. Thursday will work great! I'll send through a meeting invite :)
Thanks again for agreeing to meet!

Best,
Luke

From: Gunner, Richard <rgunner@ab.mpg.de>
Sent: Friday, October 11, 2024 5:49 PM
To: Luke Jessup <luke.jessup@uq.edu.au>
Cc: Christofer Clemente <cclement@usc.edu.au>
Subject: Re: Questions re. dead-reckoning paper

Hi Luke,

I got a bit confused with the time difference—I believe you're 'only' 8 hours ahead of me. How about we aim for next Thursday morning my time (Thursday evening your time)? Would 9 AM my time (5 PM your time) work for you?

Best,

Rich

From: Luke Jessup <luke.jessup@uq.edu.au>
Sent: 10 October 2024 13:29:58
To: Gunner, Richard
Cc: Christofer Clemente
Subject: Re: Questions re. dead-reckoning paper

Hi Rich,

Thank you for your reply and for agreeing to meet!
Next Thu or Fri evening your time (Fri or Sat morning my time) would work! Are you based in Germany at the moment? If you'd prefer evenings over mornings / early afternoon, I can make

4am or 5am work (8pm or 9pm your time). Let me know what suits best 😊

Luke

On 9 Oct 2024, at 6:50 pm, Gunner, Richard <rgunner@ab.mpg.de> wrote:

Hi Luke,

That's no problem at all! I'm flexible for a chat this Friday or next Thursday/Friday. Let me know what times work best for you. Considering the time difference, I can be flexible in the evenings (your mornings).

Best,

Rich

From: Luke Jessup <luke.jessup@uq.edu.au>
Sent: 08 October 2024 08:45:23
To: Gunner, Richard
Cc: Christofer Clemente
Subject: RE: Questions re. dead-reckoning paper

Hi Rich,

I hope you've been well and had a great rest of your break!
Chris and I were wondering if we'd be able to set up a Zoom meeting with you in the next few weeks, if you have time? It'd be brilliant to get your thoughts on a few questions we have around the data rotation stuff and future device deployments.

Best wishes,
Luke

From: Luke Jessup
Sent: Thursday, September 12, 2024 7:26 PM
To: Gunner, Richard <rgunner@ab.mpg.de>
Cc: Christofer Clemente <cclement@usc.edu.au>
Subject: RE: Questions re. dead-reckoning paper

Hi Rich,

Oh I hadn't thought to do that – I'll give that a try – thanks!
I might reach out to you again when you're back from leave to organise a meeting if that's OK? It'd be awesome to get your thoughts on a few questions that Chris and I have about this stuff and future deployments.

Best wishes,
Luke

From: Gunner, Richard <rgunner@ab.mpg.de>
Sent: Wednesday, September 11, 2024 9:09 AM
To: Luke Jessup <luke.jessup@uq.edu.au>

Cc: Christofer Clemente <cclement@usc.edu.au>

Subject: Re: Questions re. dead-reckoning paper

Hi Luke,

No problem, just a quick reply for now. What I suggest is to first run the compass function without any offsets applied to derive the pitch and roll angles. Then, determine the mean values of both during periods where you believe the animal and collar are in a normal position, such as during low-activity, steady-state behavior. Once you have these values, re-run the function and input them as the 'pitch.offset' and 'roll.offset' parameters.

In the function, if offsets are provided, the magnetic and gravity vectors are de-rotated according to the supplied Euler angles. For example, if you supply a positive pitch offset value of 45°, this reflects that the device is pitched up by 45° from the level, and the function will de-rotate the supplied (smoothed) acc and mag (normalised) values by this amount. Similarly, a positive roll value indicates a rightward tilt (bank angle), and a positive yaw value represents a clockwise rotational offset

Best,

Rich

From: Luke Jessup <luke.jessup@ug.edu.au>

Sent: 11 September 2024 00:27:42

To: Gunner, Richard

Cc: Christofer Clemente

Subject: RE: Questions re. dead-reckoning paper

Hi Rich,

Thanks so much for the detailed reply and for responding while on leave! Please feel free to respond to this when you get back from vacation – hope it's been great one so far!

Specific to our case, we're working with data collected on impala. The animals were collared with the IMU (+GPS) positioned on the front of the neck and, thus, have a pitch and roll offset from flat. I've collected some pilot data in relatively controlled conditions – walking around with the device in an ~ orientation that it would be on the impala. My main concern at the moment is removing those pitch and roll offsets from the accelerometer and magnetometer data – the results that I get from the Gundog functions are perfect when I've walked around with the device oriented flat (~0 pitch, ~0 roll, z-axis ~ pointing directly up/down).

My current approach - on the rotated data - has been to smooth the accelerometer and magnetometer data using a rolling mean (as in your code), select a point corresponding to the steady-state orientation of the device, use the pitch and roll angle at that point to create rotation matrices and apply those matrices to the smoothed accelerometer and magnetometer data, and then run Gundog.Compass and Gundog.Tracks. Would you expect this general approach to work? (provided of course the settings that I'm using for Gundog.Compass and Gundog.Tracks are correct).

If you think it'll be much more complicated than this, would it be OK if we set up a meeting to discuss? Again, really appreciate all of the information you've provided thus far, and definitely happy to provide you with feedback on the Gundog functions once we have it all working on our end – we're hoping to extend the approach to kangaroos which will make as interesting use case.

Best wishes,
Luke

From: Gunner, Richard <rgunner@ab.mpg.de>
Sent: Wednesday, September 11, 2024 6:25 AM
To: Luke Jessup <luke.jessup@uq.edu.au>
Cc: Christofer Clemente <cclement@usc.edu.au>
Subject: Re: Questions re. dead-reckoning paper

Hi Luke,

Apologies for the delayed response; I am currently on an extended vacation (post-fieldwork recovery mode 😊).

Regarding your question, I assume you're looking to derive heading from magnetometer and acceleration values prior to dead reckoning. If you're using the Gundog.Compass script from my GitHub (and I'm currently working on updates to the Gundog.Tracks script with some improvements, so feel free to share any feedback or questions), the key is to focus on identifying the portion of the data where you performed magnetometer calibration and to ensure you know how your x, y, z accelerometer and magnetometer channels are aligned with the animal's surge, heave, and sway axes, for your specified tag coordinate system within the function.

Just in case it helps I elaborate background of heading computation and coordinate frame issues below:

When there is no dynamic acceleration, the x, y, and z accelerometer values reflect the device's inclination relative to Earth's gravity vector. Gravity-based (static) acceleration can be separated from the raw values using a low/high-pass filter— I usually use a simple running mean. I typically use a window width of about 1 second. This filtering is not performed within the compass function, so beforehand, I recommend calculating smoothed accelerometer and magnetometer values. For 40 Hz data, I would use a smoothing window of around 40 for accelerometers and 10 for magnetometers (though this is completely arbitrary and you may prefer to experiment). You would then supply these smoothed values into the function.

Within the function, pitch and roll angles are approximated from the supplied smoothed accelerometer values (which are normalized within the function). Since accelerometers are insensitive to rotations about the gravity vector (heave axis), heading is computed using the arctangent of the ratio of the two horizontal magnetic field components. For accurate heading computation, these components must be aligned parallel to the Earth's surface. However, this alignment can be disrupted due to initial tag placement (e.g., on a collar, which may roll) and the animal's various postures as it moves across different terrains. To correct for this, the magnetic vectors need to be 'de-rotated' based on the pitch and roll angles of the device.

Magnetometer calibration occurs within the compass function using data from the calibration period, and the correction factors derived from this calibration are applied to all the magnetometer data. However, the function does not rotate the magnetometer or accelerometer data during ellipsoid fitting or while addressing hard iron offsets. If your tag has a consistent minor offset (e.g., due to collar placement with a slight tilt), and you're unsure about the exact amount, it's usually not necessary to worry about correcting for this. The pitch and roll values from the device will reflect its orientation relative to the ground (gravity vector) and will automatically correct the calibrated magnetometer vectors before calculating the heading. If the collar rolls smoothly and independently of the animal, the same principle applies.

The bigger issue arises if the tag is significantly offset beyond your specified coordinate frame orientation, potentially leading to gimbal lock and diminished horizontal magnetic field components. This

occurs when the pitch angle approaches $\pm 90^\circ$, causing the device's axis to align with the gravity vector. At this point, one degree of freedom is lost, making it difficult to compute accurate heading (yaw), as rotations around the yaw and roll axes become ambiguous. If you're using quaternions instead of Euler angles (as in the 'SAAM' method in the compass function), gimbal lock is avoided, since quaternions allow for smooth rotations without singularities. However, quaternions won't solve the issue when the sensor is close to vertical: the horizontal components of the magnetic field (used for heading calculation) become less accurate because the magnetic field, which would typically project onto the horizontal plane, now projects more onto the vertical axis, reducing the magnitude of the horizontal components and making them less reliable for heading calculation.

It's also worth considering the animal in question. For example, if the tag is offset on the animal—pitched up by 45 degrees from the animal's natural coordinate frame—and the animal moves in a way that causes the device to rotate an additional 60 degrees, the tag might end up inverted. In this case, without correcting for the offset, the system could incorrectly assume the animal is heading in the opposite direction. If I tag a penguin on its lower back, its posture changes drastically between walking and swimming. When the penguin is walking, the tag will be nearly vertical, while during swimming, it will be more horizontal. In both cases, the tag's coordinate frame relative to the penguin's natural movement (and the ground or water surface) will be different. To ensure accurate heading calculations, I would need to define different tag coordinate systems for walking and swimming and compute the heading separately for each period—on land and at sea—otherwise, my dead-reckoned (DR) tracks could be very inaccurate.

If you know there's a specific yaw offset in the tag placement, you can specify it in the compass function. If you're unsure, you could compare the heading correction factors from Gundog.Tracks. If the headings are consistently biased in one direction, it could indicate a tag yaw offset or the need for magnetic declination adjustment. But again, if the tag's yaw offset is minor and consistent, it's not essential to correct this, as the dead-reckoned track will be consistently biased (in theory, the patterns and scales of tortuosity between GPS points should remain the same, just directionally offset somewhat), and the track segments get rotated when corrected with their endpoint GPS fixes during the correction process anyway.

Let me know if you need further clarification

Hope this helps,

Best,

Rich

From: Luke Jessup <luke.jessup@ug.edu.au>
Sent: 10 September 2024 05:15:37
To: Gunner, Richard
Cc: Christofer Clemente
Subject: RE: Questions re. dead-reckoning paper

Hi Richard,

Just following on. I have rotation matrices that, from what I can deduce, do a good job of transforming the acceleration data but less so the magnetometer data. I have a couple of questions from the 'Preparing the three axes of rotation for derivation of heading' section of your paper (sorry if I didn't pick up on these in the text!). First, I was wondering where in the process you recommend performing the transformation? (i.e. after smoothing the raw data? before running Gundog.Compass?). Also, if it is before running Gundog.Compass, how would you go about estimating yaw angle to use in your rotation matrices?

Thanks again and apologies for the multiple emails!

Luke

-----Original Message-----

From: Luke Jessup

Sent: Tuesday, September 3, 2024 10:38 AM

To: Gunner, Richard <rgunner@ab.mpg.de>

Cc: Christofer Clemente <cclement@usc.edu.au>

Subject: RE: Questions re. dead-reckoning paper

Hi Richard,

I have another question re. your dead-reckoning paper if that's OK. So we collected data with the IMU fixed within a collar around the animal's neck - the IMU is on an angle relative to the animal frame and earth frame. Am I correct that we have to re-orient the data using rotation matrices prior to using Gundog.Compass and Gundog.Tracks? Or is there some in-built functionality for this?
So far the code works great!

Thank you,
Luke

-----Original Message-----

From: Luke Jessup

Sent: Friday, June 14, 2024 9:18 AM

To: Gunner, Richard <rgunner@ab.mpg.de>

Subject: RE: Questions re. dead-reckoning paper

Hi Richard,

Thanks so much for the swift reply, especially given limited internet.
That answers those questions perfectly. It's a great bit of code, and as a matlab user, new to R, it's been really easy to follow. I'll let you know if we come up with any further questions.
Enjoy Panama - sounds fun!

Best,
Luke

-----Original Message-----

From: Gunner, Richard <rgunner@ab.mpg.de>

Sent: Friday, June 14, 2024 8:09 AM

To: Luke Jessup <luke.jessup@uq.edu.au>

Subject: Re: Questions re. dead-reckoning paper

Hi Luke,

I'm currently in a small island in Panama at the moment so limited internet, so quick response now, but I'm happy to talk more next week onwards if you have any further questions... kangaroo deadreckoning certainly sounds fun!

Yes, one good mag cal (ensure not wearing watches /have phones/any magnetic interfering stuff on person etc. nearby when doing this) before deployment is good enough... certainly doesn't hurt to do another after if tag is still running after retrieval. No need to do another during.

In terms of movement bouts, the simple approach would be to use some sort of acceleration magnitude threshold

(e.g. VeDBA)... but for example, something like a kangaroo should be very obvious with jumping behaviour that should come out in acceleration traces as clear stylised patterns which could be manually discerned or via machine learning if you were able to do some groundtruthing of behaviours during the data acquisition period.

Best,

Rich

From: Luke Jessup <luke.jessup@uq.edu.au>

Sent: 13 June 2024 11:38

To: Gunner, Richard

Subject: Questions re. dead-reckoning paper

Dear Dr. Gunner,

I hope this email finds you well.

I'm with the animal biomechanics and biorobotics lab at the University of the Sunshine Coast, and I had a few questions re. your Dead-reckoning animal movements in R paper. We're collecting some GPS-IMU data on impala and kangaroos and were hoping to implement your dead-reckoning code.

I was wondering about the magnetometer calibration periods; how often would you recommend performing these calibrations, assuming the devices are deployed for 5-10 days at a time? Do you think that performing a calibration immediately before and after deployment would suffice? Or would you recommend during the deployment as well? Also, I was curious what approach your group took to detect bouts of movement and when the animal is stationary?

Thank you for your time!

Best wishes,

Luke

Luke Jessup

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